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#include <stdio.h>

#include <stdlib.h>

#define MAX 20 // Maximum number of vertices

// Queue implementation

struct Queue {
    int items[MAX];
    int front, rear;
};

// Initialize queue

void initQueue(struct Queue* q) {
    q->front = -1;
    q->rear = -1;
}

// Check if queue is empty

int isEmpty(struct Queue* q) {
    return (q->front == -1);
}

// Enqueue operation

void enqueue(struct Queue* q, int value) {
    if (q->rear == MAX - 1) {
        printf("Queue Overflow!\n");
        return;
    }
    if (q->front == -1) q->front = 0;
    q->items[++q->rear] = value;
}

// Dequeue operation

int dequeue(struct Queue* q) {
    if (isEmpty(q)) {
        printf("Queue Underflow!\n");
        return -1;
    }

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    }

    int item = q->items[q->front];
    if (q->front >= q->rear) {
        q->front = q->rear = -1;
    } else {
        q->front++;
    }

    return item;
}

// BFS traversal
void BFS(int adj[MAX][MAX], int visited[MAX], int start, int n) {
    struct Queue q;
    initQueue(&q);
    visited[start] = 1;
    enqueue(&q, start);
    printf("BFS Traversal starting from vertex %d: ", start);
    while (!isEmpty(&q)) {
        int current = dequeue(&q);
        printf("%d ", current);
        // Visit all adjacent vertices
        for (int i = 0; i < n; i++) {
            if (adj[current][i] == 1 && !visited[i]) {
                visited[i] = 1;
                enqueue(&q, i);
            }
        }
    }

    printf("\n");
}

// Main function
int main() {

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int n; // number of vertices

int adj[MAX][MAX]; // adjacency matrix

int visited[MAX] = {0};

int start;

printf("Enter number of vertices: ");

scanf("%d", &n);

printf("Enter adjacency matrix (%d x %d):\n", n, n);

for (int i = 0; i < n; i++) {

    for (int j = 0; j < n; j++) {

        scanf("%d", &adj[i][j]);

    }

}

printf("Enter starting vertex (0 to %d): ", n - 1);

scanf("%d", &start);

BFS(adj, visited, start, n);

return 0;

}

```

OUTPUT:

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Enter number of vertices: 5
Enter adjacency matrix (5 x 5):
0 1 1 0 0
1 0 0 1 1
1 1 0 1 0
0 0 1 1 0
0 1 0 1 0
Enter starting vertex (0 to 4): 0
BFS Traversal starting from vertex 0: 0 1 2 3 4

Process returned 0 (0x0)    execution time : 51.137 s
Press any key to continue.

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